

Data Collection and preprocessing

The first stage of this project focused on compiling a comprehensive occurrence dataset for the American pika (*Ochotona princeps*). Particular emphasis was placed on collecting not only presence records but also reliable absence records. In this context, absence refers to locations where pikas had previously been recorded but were not detected during the most recent field surveys. Each occurrence record contained the geographic coordinates, year of observation, and species status, where 1 represents a presence and 0 represents an absence.

Once the occurrence records had been assembled, climatic and topographic information was extracted for every observation location. Topographic variables, including elevation and slope, were derived directly from digital elevation data. Climatic predictors were obtained from two complementary sources. WorldClim v2.1 provides long-term climate normals (1970–2000) and therefore only required the geographic coordinates to retrieve the corresponding bioclimatic variables. In contrast, TerraClimate provides monthly climate data, so the year of each observation was matched with the corresponding annual climate record to obtain climate conditions representative of the observation period.

Following data extraction, the dataset underwent several preprocessing steps. Records containing missing values were removed to ensure that all observations used for model training contained complete environmental information. Duplicate records were also identified and removed to avoid biasing the learning process. After cleaning, the final dataset consisted of 14,393 presence records and 187 absence records. Although the dataset exhibits a substantial class imbalance, this issue was addressed during the model training and evaluation stages through appropriate sampling and model validation strategies.

Before training the model, variables that could introduce unwanted spatial or temporal bias were excluded. Specifically, the geographic coordinates and year of observation were removed from the predictor set. While these variables are essential for extracting environmental information, including them as predictors could allow the model to learn the locations or time periods represented in the training data rather than the underlying environmental conditions that determine habitat suitability. The final dataset therefore consisted exclusively of environmental predictors describing topography, long-term climate, temperature extremes, drought conditions, and snow dynamics.

Initial Features Engineering

Prior to model development, an initial set of **35 environmental predictor variables** was selected based on their ecological relevance to the American pika (*Ochotona princeps*). The selection was informed by a comprehensive review of the scientific literature on species distribution modelling and American pika ecology, with particular emphasis on the climatic and environmental factors known to influence the species' distribution and survival.

The predictor set combines long-term climatic variables from **WorldClim v2.1**, topographic variables, and climate indicators derived from **TerraClimate**. The WorldClim bioclimatic variables were included in their original form, as they represent long-term average climatic conditions that define the species' climatic niche. In contrast, the monthly TerraClimate data were aggregated into ecologically meaningful indicators, such as seasonal temperature

extremes, drought severity, snow-related metrics, and seasonal averages. These aggregated variables were designed to better represent the environmental conditions that directly affect pika physiology, forage availability, winter insulation, and overall habitat suitability.

The resulting set of predictors provides a comprehensive description of the topographic, climatic, hydrological, and snow-related factors that are expected to influence the occurrence of the American pika. A summary of the environmental predictor variables and their ecological significance is presented below.

Environmental Predictor Variables

The final modelling dataset contained the following environmental predictors.

Variable	Ecological relevance for habitat suitability
Elevation (m)	Elevation strongly influences temperature, snow persistence, vegetation communities, and oxygen availability. American pikas are predominantly alpine species and are typically associated with higher elevations where cooler climatic conditions prevail.
Slope (°)	Steeper slopes are more likely to contain talus fields, the preferred habitat of pikas, providing shelter, nesting sites, and thermally buffered microhabitats.
BIO1 – Annual Mean Temperature (Celsius)	Represents the average thermal conditions experienced throughout the year. Pikas are highly sensitive to warming, making this one of the most important climatic predictors.
BIO4 – Temperature Seasonality	Measures the annual variability in temperature. Seasonal fluctuations influence winter snow cover, growing season length, and thermal stress.
BIO5 – Maximum Temperature of the Warmest Month (Celsius)	Captures exposure to extreme summer heat, a major limiting factor for pika survival because of their low tolerance to high temperatures.
BIO6 – Minimum Temperature of the Coldest Month (Celsius)	Represents winter temperature extremes, which influence survival and the protective role of snow insulation.
BIO12 – Annual Precipitation(mm)	Reflects the overall water availability controlling vegetation productivity and forage availability.
BIO15 – Precipitation Seasonality	Describes how evenly precipitation is distributed throughout the year, affecting plant growth patterns and snow accumulation.

Estimated Snow (mm)	Indicates total snow accumulation, which provides insulation during winter and influences water availability after snowmelt (calculated annually?).
Total Winter Precipitation (mm)	Represents winter moisture input, an important determinant of snowpack formation (calculated annually?).
Months with Snow	Indicates the duration of snow cover, affecting winter insulation and the timing of vegetation growth. (calculated annually)
Months Above 25°C	Measures the frequency of potentially stressful high-temperature conditions that may limit pika activity and survival (calculated annually).
Maximum Temperature (Celsius)	Represents annual heat extremes experienced during the year of observation (calculated annually).
Winter Maximum Temperature (Celsius)	Indicates unusually warm winter conditions that may reduce snow persistence (calculated annually).
Summer Minimum Temperature (Celsius)	Reflects cool nighttime temperatures during the growing season.(calculated annually).
Winter Minimum Temperature(Celsius)	Captures the harshest winter conditions experienced during the observation year.
Worst Drought (PDSI)	(Minimum of the Palmer Drought Severity Index). Represents the most severe drought conditions during the observation year, influencing vegetation productivity.
Worst Drought – Previous Year	Accounts for lagged ecological effects of drought on vegetation and habitat quality.
Annual Mean PDSI	(Average of the Palmer Drought Severity Index)Measures the average moisture availability during the observation year.
Annual Mean PDSI – Previous Year	Captures delayed effects of long-term moisture conditions on ecosystem productivity.
Summer Mean PDSI	Represents moisture availability during the primary growing season when haypiles are collected.
Summer Mean PDSI – Previous Year	Accounts for delayed effects of summer drought on vegetation recovery.
Months with Drought	Indicates the duration of drought conditions affecting forage production (calculated annually).

Months with Drought – Previous Year	Captures cumulative drought stress from the preceding year.
Winter Mean PDSI	Represents winter moisture conditions influencing snow accumulation and soil moisture recharge.
Winter Mean PDSI – Previous Year	Accounts for lagged winter moisture effects on habitat conditions.
Maximum Snow Water Equivalent (SWE) (mm)	Measures the maximum amount of water stored in the seasonal snowpack, an important indicator of snow depth.
Maximum SWE – Previous Year (mm)	Captures delayed influences of previous snow conditions on vegetation and hydrology.
Annual Mean Snow Water Equivalent (mm)	Represents the average amount of water stored in snow throughout the year.
Annual Mean SWE – Previous Year	Reflects the carry-over effects of previous snow conditions.
Cold Season Mean SWE	Quantifies average snow water equivalent during winter, representing snowpack persistence.
Cold Season Mean SWE – Previous Year	Captures delayed winter snow effects on habitat suitability.
Spring Mean SWE	Represents snow conditions during spring, which influence snowmelt timing and the onset of vegetation growth.
Spring Mean SWE – Previous Year	Accounts for lagged effects of spring snow conditions on subsequent habitat quality and forage availability.

These predictors collectively describe the principal environmental factors influencing American pika habitat suitability. Topographic variables characterize the physical structure of alpine habitats, while climatic variables capture both long-term climatic conditions and interannual variability. Temperature-related predictors quantify thermal stress, precipitation and drought variables describe water availability and vegetation productivity, and snow-related variables characterize winter insulation and seasonal hydrological processes. Together, they provide a comprehensive representation of the environmental conditions that determine the distribution and persistence of pika populations under both current and future climate scenarios.

Features Selection

Because our predictors consist of climate variables, it is expected that many of them are highly correlated, as they describe related aspects of the same climate system.

Consequently, the objective of feature reduction is not simply to remove correlated variables, but to identify predictors that are both highly correlated and provide little additional predictive information beyond what is already captured by the remaining variables.

To achieve this, we initially trained the model using all 35 predictor variables. We then evaluated the contribution of each predictor through a SHAP (SHapley Additive exPlanations) analysis to determine which correlated variables provided the greatest predictive value and which could be removed without compromising model performance.

SHAP analysis

SHAP (SHapley Additive exPlanations) is an explainable artificial intelligence technique that quantifies the contribution of each feature to a model's predictions. It is based on Shapley values from cooperative game theory, where each feature is considered a "player" contributing to the final prediction. By averaging the absolute SHAP values across all observations, the method ranks predictors according to their overall importance to the model.

The SHAP analysis of the testing dataset produced the initial feature importance ranking shown below (Figure 1).

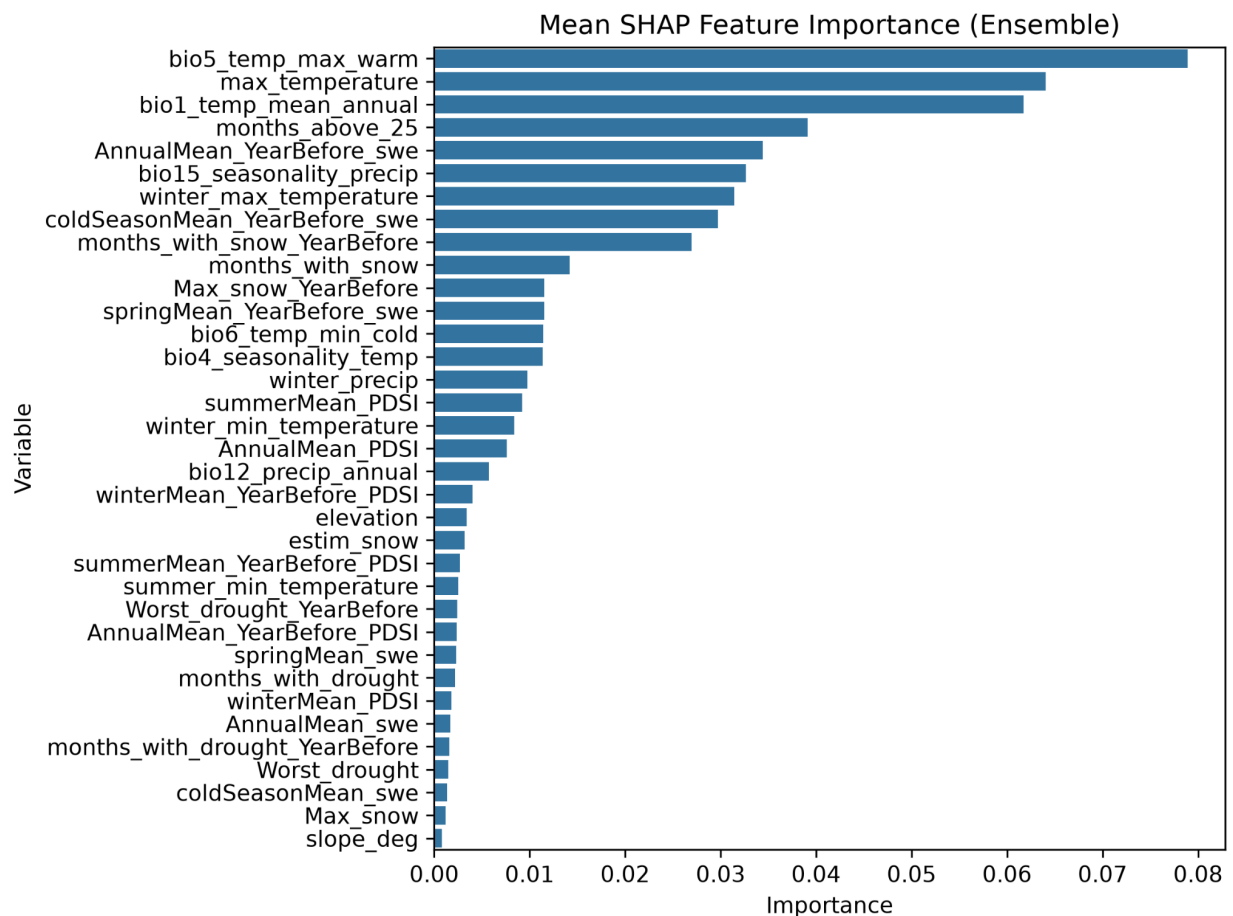


Figure 1: SHAP initial analysis for the initial set of predictors.

The results indicate that several variables like Max snow or slope contribute substantially less to the predictions than others like max temperature. Therefore, we adopted an iterative feature selection approach in which low-importance variables identified by SHAP were further evaluated using their pairwise Spearman correlation coefficients. This allowed us to remove redundant predictors while preserving the model's predictive performance, resulting in a more interpretable feature set.

Final features selection

Following the SHAP-based feature selection process, the number of predictors was reduced from 35 to 20 without any loss in model performance. Although several of the retained variables remain correlated, each provides complementary information that improves the predictive capability of the habitat suitability model. The final feature importance rankings and SHAP summary plot are presented in **Figures 2 and 3**, respectively.

The SHAP feature importance analysis (**Figure 2**) quantifies the average contribution of each predictor to the model's predictions, with higher mean absolute SHAP values indicating greater influence on habitat suitability estimates. The results show that temperature-related variables are the primary drivers of pika habitat suitability. In particular, maximum temperature of the warmest month (bio5) is the most influential predictor, followed by maximum temperature and annual mean temperature (bio1). This highlights the importance of summer heat and overall thermal conditions in determining suitable habitat, consistent with the pika's well-established sensitivity to high temperatures.

The next most influential predictors include the number of months with temperatures above 25°C, the previous year's annual mean snow water equivalent (SWE), precipitation seasonality (bio15), winter maximum temperature, and the number of snow-covered months in the previous year. These variables suggest that, alongside thermal conditions, snowpack dynamics and seasonal precipitation patterns are important determinants of habitat suitability, likely through their effects on winter insulation, soil moisture, and the maintenance of favourable microclimatic conditions.

Predictors describing snow duration, temperature seasonality and winter precipitation made smaller but still meaningful contributions to the model. In contrast, elevation exhibited relatively low importance once climatic variables were included, indicating that its predictive value is largely explained through its relationship with climate rather than an independent ecological effect.

The SHAP summary plot (**Figure 3**) complements the feature importance rankings by illustrating how individual predictor values influence model predictions. Each point represents an observation from the testing dataset, with its position along the x-axis corresponding to the SHAP value. Positive SHAP values increase the predicted habitat suitability, whereas negative values decrease it. Point colour represents the magnitude of the predictor, with red indicating high values and blue indicating low values.

The summary plot confirms that temperature is the dominant control on habitat suitability. For the three most important predictors—annual mean temperature (bio1), maximum temperature of the warmest month (bio5), and maximum temperature—higher temperatures (red points) are generally associated with negative SHAP values, indicating reduced habitat

suitability. Conversely, lower temperatures (blue points) tend to contribute positively to model predictions. This relationship is consistent with the ecological requirements of pikas, which are highly vulnerable to heat stress and depend on cool alpine environments.

Snow-related variables generally exhibit the opposite trend. Higher values of annual mean SWE from the previous year, maximum SWE, and the number of snow-covered months are generally associated with positive SHAP values, indicating that greater snow accumulation and longer-lasting snow cover increase habitat suitability. A persistent snowpack likely provides thermal insulation during winter while maintaining favourable moisture and temperature conditions throughout the year.

Precipitation seasonality (bio15) displays a more complex response, suggesting that its influence depends on the broader climatic context rather than following a simple linear relationship.

Overall, the SHAP analyses demonstrate that thermal conditions are the principal drivers of pika habitat suitability, with snowpack characteristics acting as important secondary controls. The agreement between the feature importance rankings and the SHAP summary plot indicates that the final set of 20 predictors captures the key environmental processes governing pika habitat suitability while avoiding unnecessary redundancy, resulting in a more interpretable predictive model.

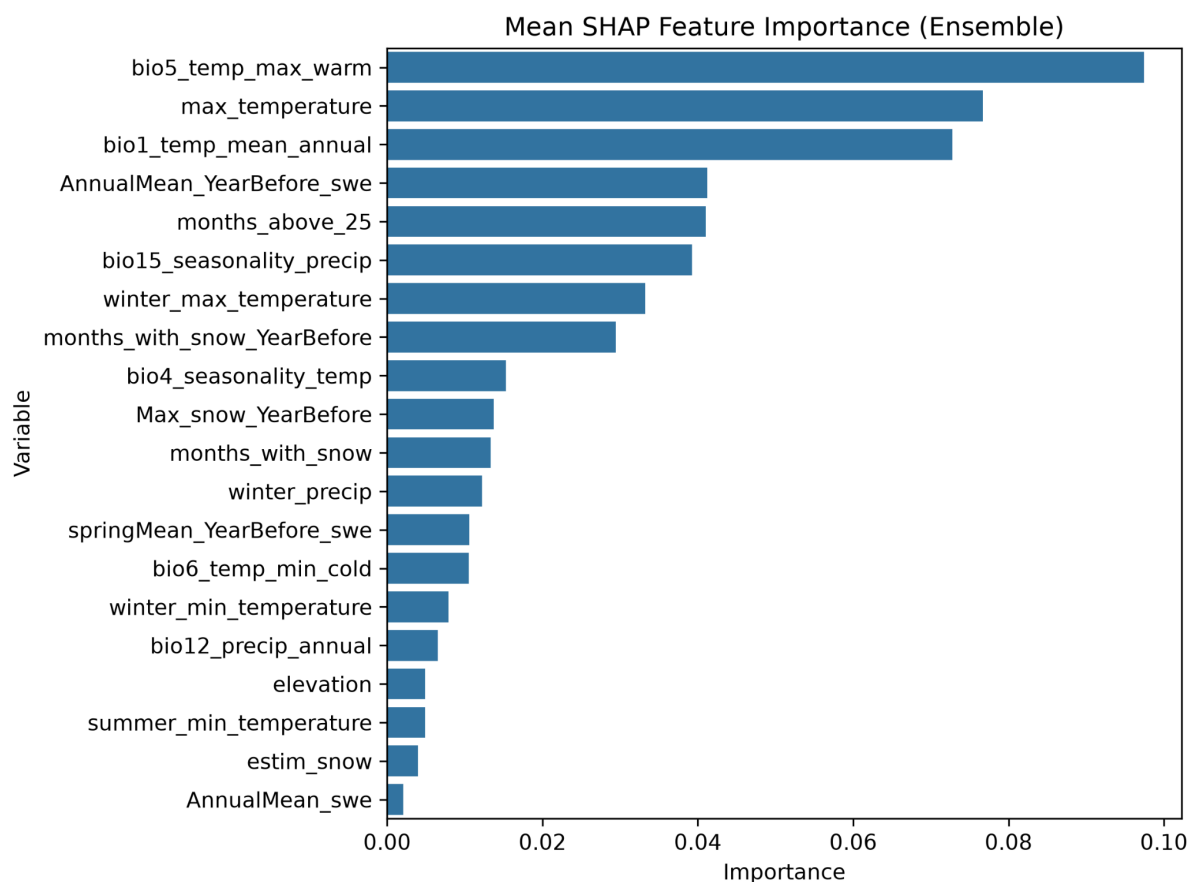


Figure 2: SHAP analysis for the final set of predictors.

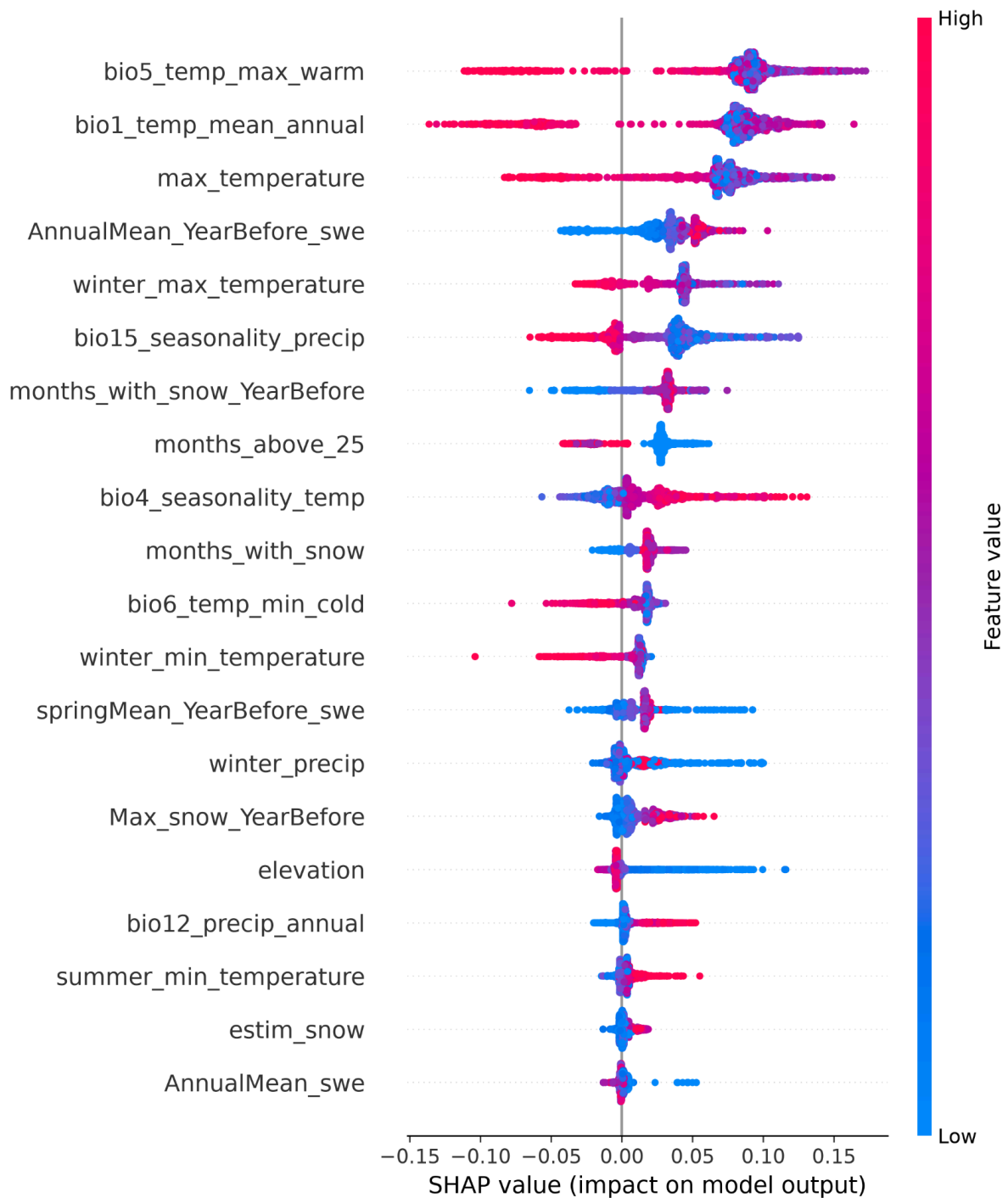


Figure 3: SHAP summary plot for the final set of predictors.

ML Models Training

The final dataset comprised 14,393 presence records and 187 absence records, resulting in a highly imbalanced dataset. Because machine learning classifiers trained on severely imbalanced data tend to become biased toward the majority class, a balanced training strategy was adopted. For each training iteration, a random subset of approximately 200–225 presence records was selected and paired with the available absence records. This

approach ensured that the models learned to discriminate between suitable and unsuitable habitat rather than simply predicting the majority class.

Prior to model training, 20% of the balanced dataset was randomly withheld as an independent testing set, while the remaining 80% was used for model development.

Three supervised classification algorithms were evaluated: Random Forest (RF), Extreme Gradient Boosting (XGBoost), and Light Gradient Boosting Machine (LightGBM).

First, for each algorithm, hyperparameters were iteratively optimized using cross-validation to identify the parameter combination that maximized predictive performance while minimizing overfitting.

Following hyperparameter optimization, an ensemble modelling approach was implemented for each algorithm. During each ensemble iteration, a new random subset of approximately 200 presence records was combined with the available absence records, again reserving 20% of the data for testing. **A total of 100 models were trained for each algorithm. The predicted probability of habitat suitability for each observation was obtained by averaging the predictions across all 100 models.** The final prediction was then converted into a binary classification, with observations assigned as presence when the mean predicted probability was ≥ 0.5 and absence otherwise.

This ensemble strategy reduces the influence of the random selection of presence records, decreases model variance, and produces more stable and robust predictions than relying on a single model trained on one balanced subset.

Results: Model Evaluation

The independent testing dataset was used to evaluate the predictive performance of the three ensemble classifiers. Although Random Forest, XGBoost, and LightGBM all achieved good performance, the Random Forest ensemble consistently produced the highest overall accuracy and was therefore selected as the final habitat suitability model(see **Figure 4** for the confusion matrix for the Random Forest ensemble model).

The predictive performance of the Random Forest ensemble was evaluated using several complementary metrics, each describing a different aspect of model performance. Together, these metrics provide a comprehensive assessment of the model's ability to distinguish between suitable (i.e. presence) and unsuitable (i.e. absence) habitat for the American pika.

The model achieved a **Receiver Operating Characteristic Area Under the Curve (ROC-AUC)** of **0.999**. ROC-AUC measures the model's ability to discriminate between presence and absence observations across all possible classification thresholds by comparing the true positive rate (sensitivity) with the false positive rate ($1 - \text{specificity}$). A value of **1.0** indicates perfect discrimination, whereas a value of **0.5** represents random classification. The ROC-AUC of 0.999 therefore indicates that the model has the ability to distinguish suitable from unsuitable habitat.

The model also achieved a **balanced accuracy** of **0.99**. Unlike conventional accuracy, balanced accuracy gives equal importance to the presence and absence classes by

averaging their respective recall (sensitivity) values. This metric is particularly useful for imbalanced datasets because it prevents the majority class from dominating the overall performance measure. A balanced accuracy of 0.99 indicates that the model classified both suitable and unsuitable habitat with very high accuracy. Similarly, the **F1-score** was **0.99**. The F1-score is the harmonic mean of precision and recall and provides a single measure that balances the model's ability to correctly identify suitable habitat while minimizing incorrect predictions.

For the **presence class**, the model achieved a **precision of 1.00**, meaning that every location predicted as suitable habitat corresponded to an observed pika presence. In other words, the model produced **no false-positive predictions**, indicating that it did not incorrectly classify unsuitable habitat as suitable. The **recall (sensitivity) was 0.98**, meaning that approximately **98% of all known presence locations** were correctly identified, with only **57 of the 2,879 testing presence observations** being classified as absence (false negatives). These results demonstrate that the model is highly effective at identifying suitable habitat.

Performance for the **absence class** should be interpreted more cautiously because only **38 absence observations** were available for testing. The model correctly classified all absence locations, resulting in an **absence recall (specificity) of 1.00**, corresponding to **38 true negatives** and **no false positives**. However, the **absence precision was 0.40**. This relatively low value should not be interpreted as poor model performance. Instead, it reflects the imbalance between presence and absence observations. When one class is very rare, precision becomes highly dependent on class prevalence, meaning that even a perfectly classified minority class can yield a relatively low precision value. The performance metrics presented above were calculated using a **classification threshold of 0.5**, whereby locations with a predicted probability of presence greater than or equal to 0.5 were classified as suitable habitat. This threshold provides a conservative model that completely eliminates false-positive predictions, ensuring that all predicted suitable habitats correspond to known pika occurrences. However, threshold selection represents a trade-off between omission errors (false negatives) and commission errors (false positives). Exploratory analyses showed that reducing the classification threshold to approximately **0.4** decreases the number of **false negatives**, thereby improving the model's ability to detect suitable habitat. Consequently, the precision of the absence class increases substantially, reaching approximately **70%**, while maintaining a high overall predictive performance. Depending on the intended application, a lower threshold may therefore be preferable. For example a higher threshold is more appropriate when a conservative estimate of suitable habitat is required.

	Predicted Presence	Predicted Absence
Actual Presence	2822 (TP)	57 (FN)
Actual Absence	0 (FP)	38 (TN)

Figure 4: Confusion matrix for the independent testing sample for the Random Forest Ensemble Model

The confusion matrix shows that the model correctly classified **2,860 of the 2,917 testing observations**, corresponding to an overall accuracy of approximately **98%**. Importantly, the model generated **no false-positive predictions**, meaning that every location predicted as suitable habitat was confirmed by an observed pika presence. The only classification errors were **57 false negatives**, representing known presence locations that were predicted as unsuitable. This indicates that, at the default probability threshold of **0.5**, the Random Forest ensemble behaves as a conservative classifier, favouring the avoidance of false positives at the expense of missing a small number of true presence locations. Such behaviour is often desirable when high confidence in predicted suitable habitat is required.

Model Limitations

The Random Forest Ensemble model demonstrated excellent predictive performance on the testing dataset. However, the geographical distribution of the presence and absence observations (**Figure 5**) highlights an important limitation of the available data. While presence records are distributed across a large portion of the study area, the absence observations are few in number (187 records) and are concentrated within a limited number of survey locations. This spatial clustering reflects the design of the field surveys.

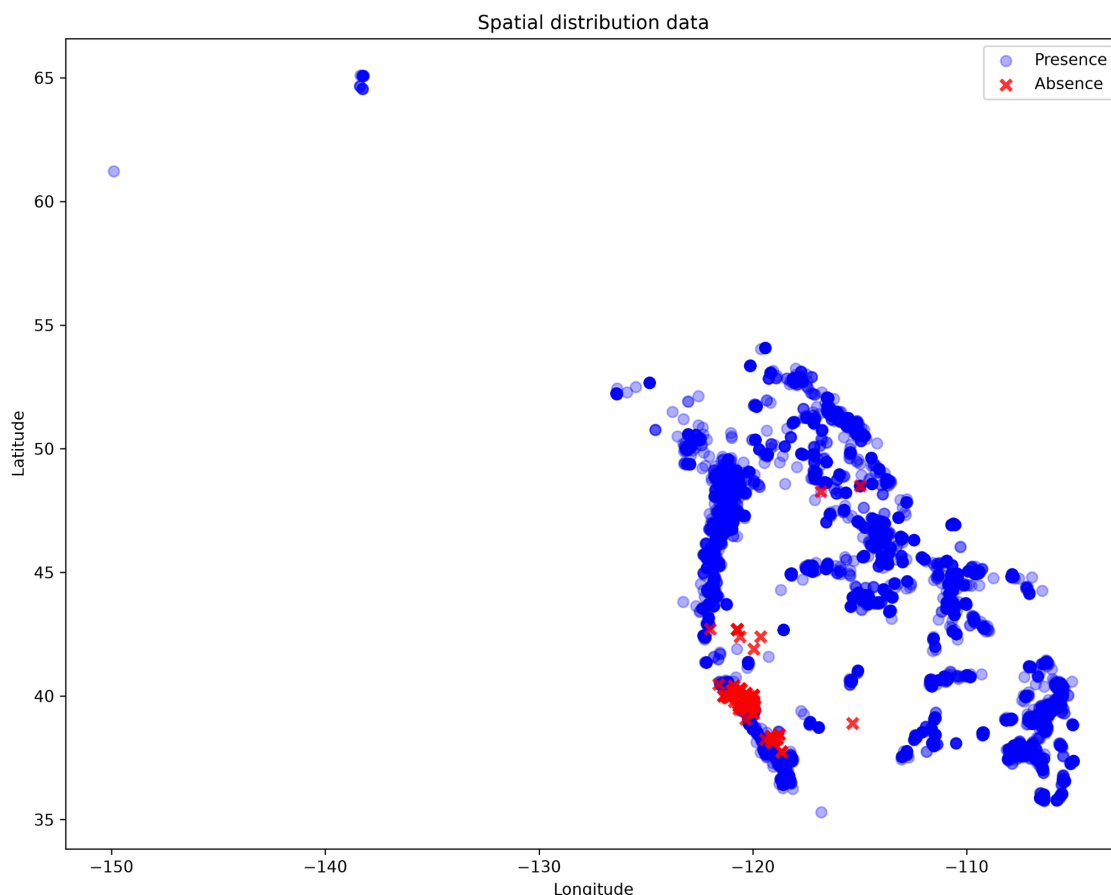


Figure 5:Geographical distribution of pika presence (blue circles) and absence (red crosses) observations used for model training and evaluation.

The limited number and clustered distribution of absence observations make it difficult to fully assess how well the model generalizes across the entire current range of the American pika. Ideally, model transferability would be evaluated using independent presence and absence data collected from geographically distinct regions. However, because absence records were scarce, nearly all available observations were required during model training to ensure a balanced dataset and maximise the model's ability to learn the distinction between suitable and unsuitable habitat.

Although a comprehensive spatial validation was therefore not possible, we assessed whether model performance varied substantially among different geographic regions represented in the dataset. The testing observations were grouped into spatial clusters based on their geographic proximity, and the Area Under the Receiver Operating Characteristic Curve (AUC) was calculated separately for each cluster. This approach provides an indication of whether the model performs consistently across different parts of the study area or whether its predictive ability is strongly influenced by location.

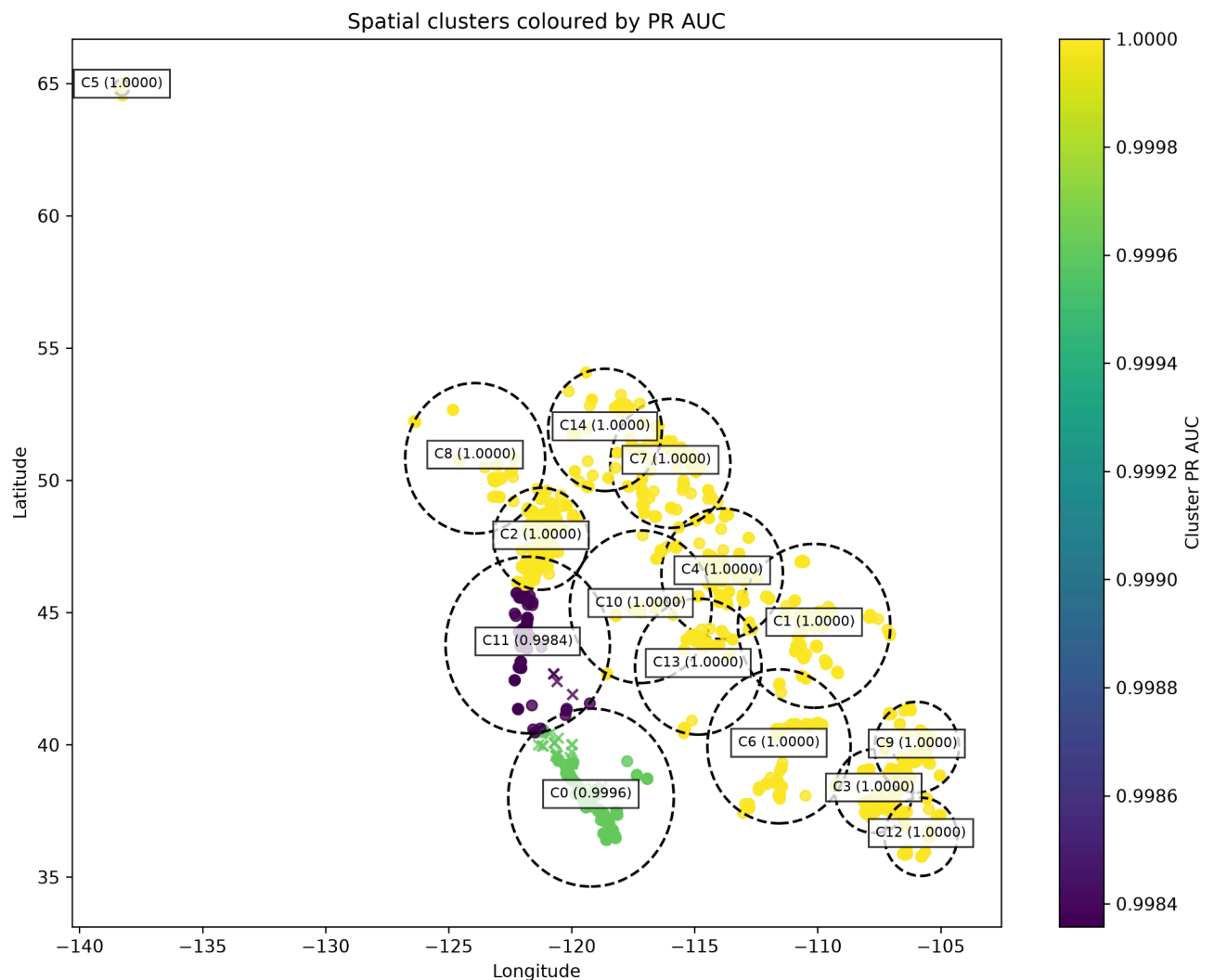


Figure 6: Area Under the Receiver Operating Characteristic Curve (ROC-AUC) calculated for each of the **15 spatial clusters** into which the testing dataset was divided. Although minor differences in predictive performance are observed among clusters, there is no evidence of substantial spatial variation, indicating that the model performs consistently across the surveyed regions.

The results (Figure 6) show that the local AUC values vary only modestly among the different spatial clusters. Although some geographic regions exhibit slightly higher or lower predictive performance than others, no substantial decrease in classification accuracy was observed. This consistency suggests that the model is not simply memorizing local environmental conditions but is capturing broader climatic relationships associated with pika habitat suitability.

Nevertheless, these results should be interpreted with appropriate caution. The spatial validation is limited to regions represented within the available survey data and does not constitute an independent test of model transferability to unsampled portions of the species' range. Consequently, the model should not be considered universally applicable across all pika habitats. Instead, the results indicate that the model can be applied with confidence to areas that are environmentally and geographically similar to those represented in the training data.

For this reason, all subsequent analyses of habitat suitability change were restricted to the geographical regions represented in the training and testing datasets. Restricting the analysis to these surveyed areas avoids extrapolating the model to regions where its predictive performance has not been evaluated and ensures that projected changes in habitat suitability are interpreted only within the environmental and geographic domain over which the model has been validated. This conservative approach increases confidence that the predicted changes reflect genuine responses to future climatic conditions rather than artefacts arising from extrapolation beyond the available observations